
EDGEVISION

PPE Compliance & Work-at-Height Safety Platform
Final Technical Submission Report

PRODUCTION MODEL — V3-HN

Report compiled from verified project documentation · August 14, 2026

SECTION 1

Executive Summary

The EdgeVision PPE Compliance Platform is an edge-deployed computer vision system designed to enforce industrial safety protocols in real time. It monitors restricted zones, verifies required PPE (helmets and vests), and applies temporal validation to keep violation alerting precise rather than noisy.

This report documents the promotion of the **V3-HN** YOLO model to production. V3-HN was trained with a hard-negative mining strategy that removed the false positives seen in earlier prototypes, while a separate **V3-BOOTS** variant remains experimental only. The system integrates a FastAPI backend, a PostgreSQL datastore, and a Next.js dashboard, and is staged — but not yet hardware-validated — for TensorRT deployment on Jetson.

SECTION 2

System Architecture

EdgeVision uses a modular pipeline that takes a camera stream through detection, tracking, association, rule evaluation, and temporal validation before a violation is logged.

Pipeline

Camera / Video Input → Person Detection & Tracking (YOLOv8n + ByteTrack) → PPE Detection (YOLOv8n, V3-HN) → Spatial Person-PPE Association → Zone / Rule Engine → Temporal Validation (hysteresis) → Violation Generation → FastAPI Backend → PostgreSQL, with results surfaced on the Next.js Frontend.

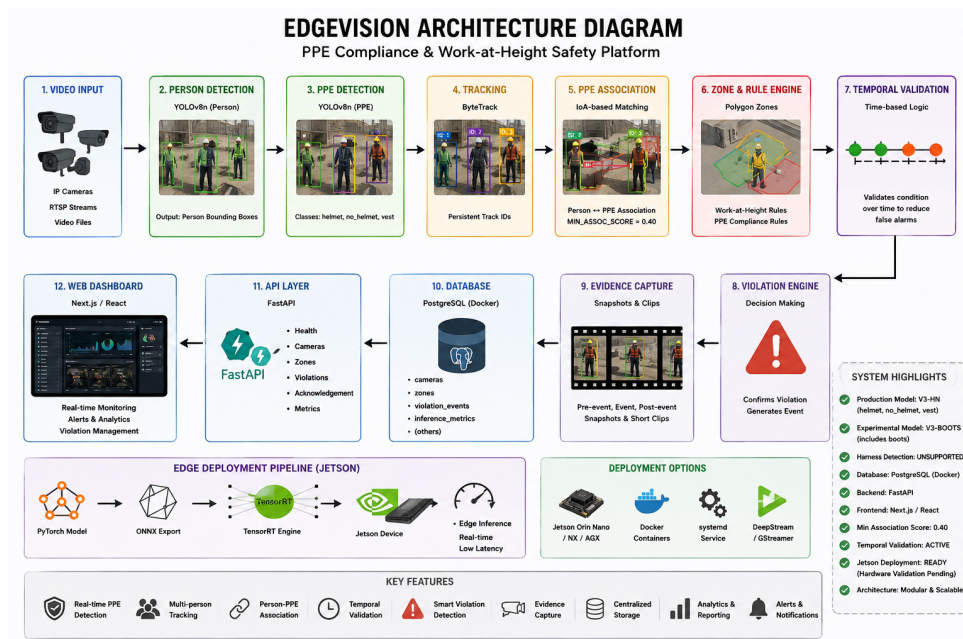


Figure 1. End-to-end system architecture, from camera input through to the web dashboard.

SECTION 3

Dataset & Hard Negatives

The system is trained on the **Construction-PPE** dataset, split 17,452 train / 2,438 validation / 2,464 test images.

During V2 testing, the model produced real-world false positives — for example, detecting yellow pipes or other yellow objects as helmets. To resolve this, the V3-HN training set was enriched with **hard negatives**: images containing visually similar objects, added with no PPE labels, so the model is explicitly taught what not to detect.

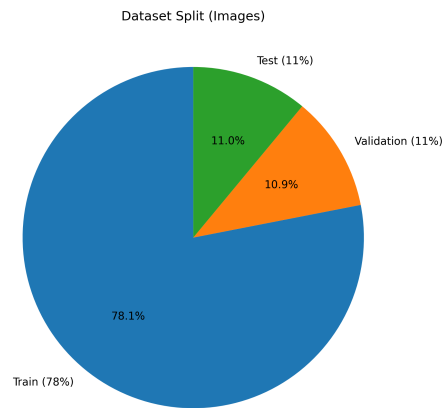


Figure 2. Image split across train / validation / test sets.

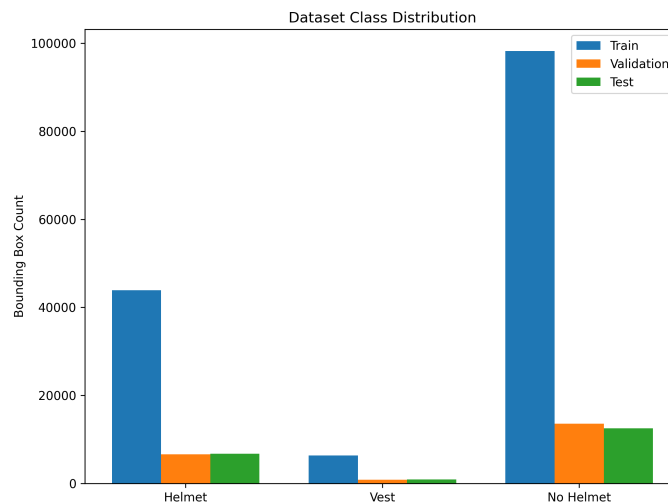


Figure 3. Bounding-box count per class across each split.

A Hard-Negative Example

Brightly colored, PPE-like imagery — such as illustrated or printed safety-wear graphics in the environment — is exactly the kind of visually similar, non-worker object the hard-negative set targets, so the model learns to distinguish a real worker's PPE from a look-alike in the background.



Figure 4. Example of PPE-like imagery relevant to hard-negative mining.

SECTION 4

Model Development & Architecture

The production model's architecture, as validated directly from its .pt checkpoint metadata:

Property	Value
Base architecture	YOLOv8n (Nano)
Layers	130
Parameters	3,012,018
Gradients	0 (inference / frozen)
GFLOPs	8.2
Classes (schema)	6 — person, helmet, no_helmet, vest, boots, harness

The 6-class schema is defined at the architecture level; boots and harness are not currently populated with sufficient training data and are not part of the validated production output (see Section 5).

Trade-off Decision

V3-HN trades a modest reduction in helmet/vest recall for substantially improved real-world false-positive behavior and improved no-helmet recall — considered a critical requirement for production trust in a safety system.

SECTION 5

Model Decision & Performance

The metric comparison used to select V3-HN for production and classify V3-BOOTS as experimental:

Metric	V2	V3-HN	V3-BOOTS
Role	Rollback	Production	Experimental
mAP50	84.45%	84.20%	80.10%
mAP50-95	50.12%	48.77%	N/A
Helmet AP50	90.15%	88.65%	82.20%
Vest AP50	82.37%	79.77%	86.10%
Boots AP50	N/A	N/A	80.10%
Boots Recall	N/A	N/A	75.30%
Warm FPS	24.3	16.2	27.82
Helmet FP	154	0	N/A
Vest FP	10	0	N/A

V3-BOOTS is experimental only and is NOT the production model. V3-HN remains production. The boots model suffers from bounding-box inconsistencies.

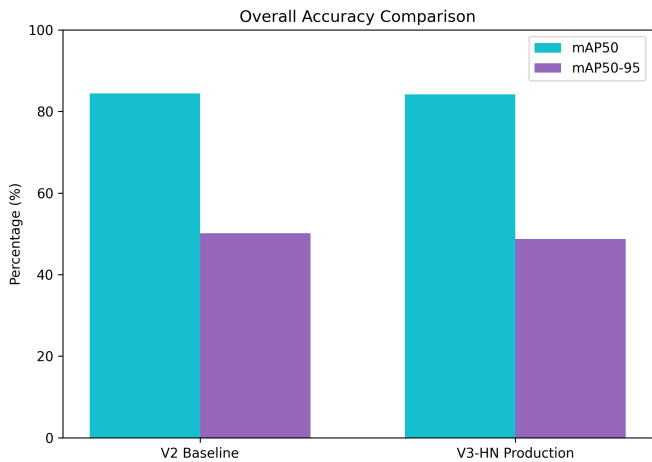


Figure 5. mAP50 and mAP50-95, V2 vs. V3-HN.

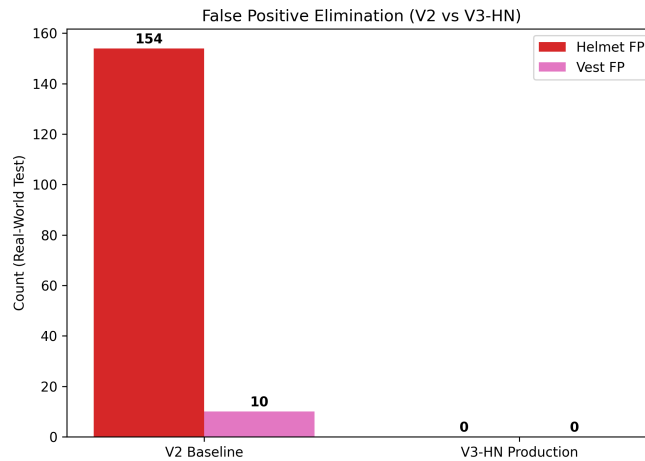


Figure 6. Real-world false positives on live test video — eliminated entirely in V3-HN.

SECTION 6

Jetson Claims & Performance

It is critical to separate development benchmarking from physical edge benchmarking. The 16.2 FPS figure reported for V3-HN was measured on an **RTX development workstation**. This does **not** prove target Jetson hardware performance.

Benchmark	Result
RTX development workstation	16.2 FPS (PASS vs. 12 FPS minimum)
Jetson target hardware	PENDING

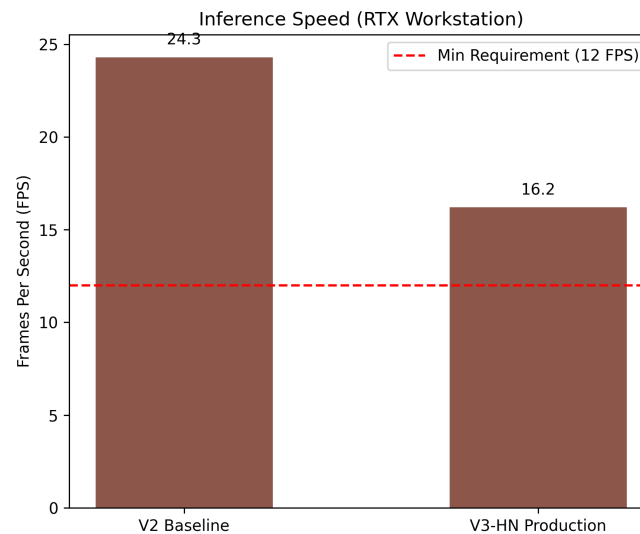


Figure 7. Inference speed on the RTX development workstation, against the 12 FPS minimum requirement.

SECTION 7

Deployment Status

The ONNX/TensorRT deployment pipeline status, tracked accurately to prevent premature claims of hardware validation:

Item	Status
ONNX export script	READY
ONNX artifact validation	PENDING
TensorRT instructions	READY
TensorRT engine validation	PENDING
Jetson physical validation	PENDING

SECTION 8

Violation Engine & Real Evidence

The violation engine correlates spatial person–PPE overlap with zone rules and temporal validation before a violation is confirmed and logged with photographic evidence.



Figure 8. Real-world evidence frame: association lines link each detected worker to their corresponding PPE detection region.

SECTION 9

Web Application & Rollback

The Next.js dashboard provides live monitoring, violation review, and safety reporting.

Values shown in dashboard screenshots (e.g. live FPS, violation counts, compliance rate) are **UI demonstration data** for interface walkthroughs — they are not production benchmark results and should not be cited as model performance figures.

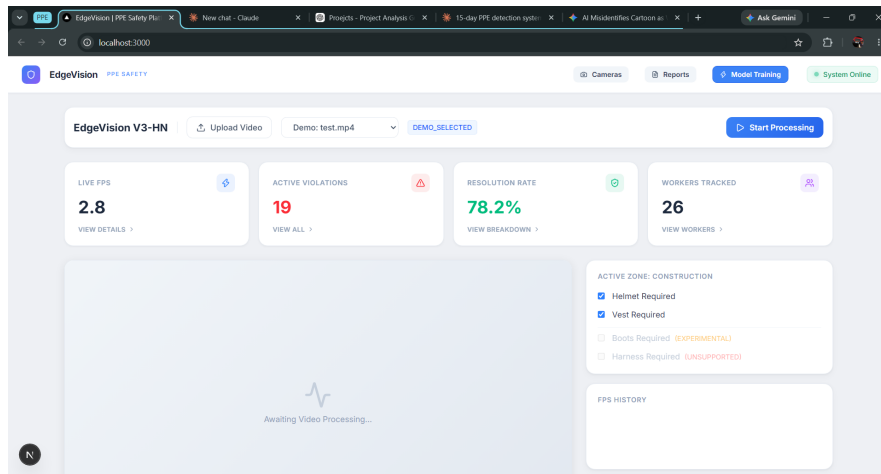


Figure 9. Dashboard overview (UI demonstration data).

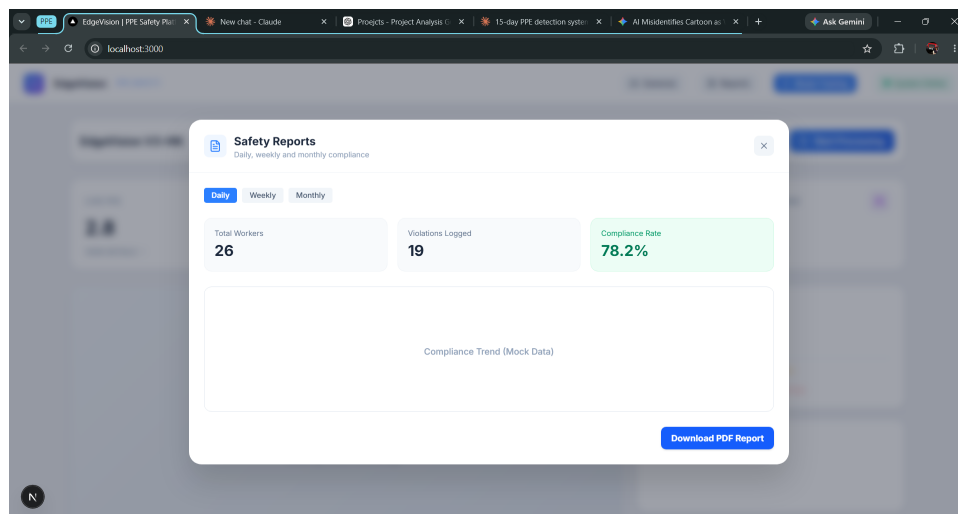


Figure 10. Safety Reports panel (UI demonstration data).

Rollback Path

If the V3-HN model encounters regressions in production, the system falls back to the preserved V2 checkpoint at:

`02_MODELS/rollback/ppe_v2_backup.pt`